

☑ Coffee Sales Analysis

```
1 import numpy as np          #NUMPY IS USED FOR NUMERICAL OPERATIONS
2 import pandas as pd # PANDAS FOR IMPORTING THE DATASET
3 import matplotlib.pyplot as plt      # FOR DATA VISUALIZATION PURPOSE
4 import seaborn as sns          # FOR DATA VISUALIZATION
5
6
```

LOADING THE DATASET:

```
1 df = pd.read_csv("/content/index.csv")      # using the Pandas library's read_csv function. The dataset is s
2 df
```

	date		datetime	cash_type	card	money	coffee_name
0	2024-03-01	2024-03-01	10:15:50.520	card	ANON-0000-0000-0001	38.70	Latte
1	2024-03-01	2024-03-01	12:19:22.539	card	ANON-0000-0000-0002	38.70	Hot Chocolate
2	2024-03-01	2024-03-01	12:20:18.089	card	ANON-0000-0000-0002	38.70	Hot Chocolate
3	2024-03-01	2024-03-01	13:46:33.006	card	ANON-0000-0000-0003	28.90	Americano
4	2024-03-01	2024-03-01	13:48:14.626	card	ANON-0000-0000-0004	38.70	Latte
...	...	...	...	...	...	...	...
1128	2024-07-31	2024-07-31	20:53:35.077	card	ANON-0000-0000-0443	23.02	Cortado
1129	2024-07-31	2024-07-31	20:59:25.013	card	ANON-0000-0000-0040	27.92	Americano with Milk
1130	2024-07-31	2024-07-31	21:26:26.000	card	ANON-0000-0000-0444	32.82	Latte
1131	2024-07-31	2024-07-31	21:54:11.824	card	ANON-0000-0000-0445	32.82	Latte
1132	2024-07-31	2024-07-31	21:55:16.570	card	ANON-0000-0000-0446	32.82	Latte

1133 rows × 6 columns

Next steps:

Generate code with df

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```
1 df.head(10)      #to check the first 10 values of the dataset
```

	date		datetime	cash_type		card	money	coffee_name
0	2024-03-01	2024-03-01	10:15:50.520	card	ANON-0000-0000-0001	38.7	Latte	
1	2024-03-01	2024-03-01	12:19:22.539	card	ANON-0000-0000-0002	38.7	Hot Chocolate	
2	2024-03-01	2024-03-01	12:20:18.089	card	ANON-0000-0000-0002	38.7	Hot Chocolate	
3	2024-03-01	2024-03-01	13:46:33.006	card	ANON-0000-0000-0003	28.9	Americano	
4	2024-03-01	2024-03-01	13:48:14.626	card	ANON-0000-0000-0004	38.7	Latte	
5	2024-03-01	2024-03-01	15:39:47.726	card	ANON-0000-0000-0005	33.8	Americano with Milk	
6	2024-03-01	2024-03-01	16:19:02.756	card	ANON-0000-0000-0006	38.7	Hot Chocolate	
7	2024-03-01	2024-03-01	18:39:03.580	card	ANON-0000-0000-0007	33.8	Americano with Milk	
8	2024-03-01	2024-03-01	19:22:01.762	card	ANON-0000-0000-0008	38.7	Cocoa	
9	2024-03-01	2024-03-01	19:23:15.887	card	ANON-0000-0000-0008	33.8	Americano with Milk	

Next steps:

Generate code with df

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```
1 df.info()          #to get the information about the columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1133 entries, 0 to 1132
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype

```

```

---
0  date      1133 non-null  object
1  datetime  1133 non-null  object
2  cash_type 1133 non-null  object
3  card      1044 non-null  object
4  money     1133 non-null  float64
5  coffee_name 1133 non-null object
dtypes: float64(1), object(5)
memory usage: 53.2+ KB

```

```
1 df.shape
```

```
(1133, 6)
```

```
1 df.describe() #to get the statistics of numerical column
```

```

money
count 1133.000000
mean   33.105808
std     5.035366
min     18.120000
25%     28.900000
50%     32.820000
75%     37.720000
max     40.000000

```

```
1 df.isnull().sum() #to check if there are null values and to sum it all
```

```

0
date      0
datetime  0
cash_type  0
card      89
money     0
coffee_name 0

```

```
1 df.columns #columns names
```

```
Index(['date', 'datetime', 'cash_type', 'card', 'money', 'coffee_name'], dtype='object')
```

```
1 df['date'] #to get the values of the date column
```



	date
0	2024-03-01
1	2024-03-01
2	2024-03-01
3	2024-03-01
4	2024-03-01
...	...
1128	2024-07-31
1129	2024-07-31
1130	2024-07-31
1131	2024-07-31
1132	2024-07-31

1133 rows × 1 columns

“ ”

1 df.dtypes



	0
date	object
datetime	object
cash_type	object
card	object
money	float64
coffee_name	object

“ ”

```
1 df['date'] = pd.to_datetime(df['date'])
```

1 df.dtypes



	0
date	datetime64[ns]
datetime	object
cash_type	object
card	object
money	float64
coffee_name	object

“ ”

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```
1 df['coffee_name'].value_counts(normalize=True).sort_values(ascending=False).round(4)*100
```



proportion	
coffee_name	
Americano with Milk	23.65
Latte	21.45
Cappuccino	17.30
Americano	14.92
Cortado	8.74
Hot Chocolate	6.53
Espresso	4.32
Cocoa	3.09



- In this we can see the most popular products which are arranged in descending order
- Americano with Milk is at the top where as the second place belongs to Latte followed by other coffees

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```
1 #Convert date and datetime to datetme format
2 df['date']=pd.to_datetime(df['date'])
3 df['datetime']=pd.to_datetime(df['datetime'])
4
5
6
```

```
1 #Create column of Month, Weekdays, and Hours
2 df['month']=df['date'].dt.strftime('%Y-%m')
3 df['day']=df['date'].dt.strftime('%w')
4 df['hour']=df['datetime'].dt.strftime('%H')
```

1 df.info()



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1133 entries, 0 to 1132
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0    date        1133 non-null   datetime64[ns]
1    datetime    1133 non-null   datetime64[ns]
2    cash_type   1133 non-null   object
3    card        1044 non-null   object
4    money       1133 non-null   float64
5    coffee_name 1133 non-null   object
6    month       1133 non-null   object
7    day         1133 non-null   object
8    hour        1133 non-null   object
dtypes: datetime64[ns](2), float64(1), object(6)
memory usage: 79.8+ KB
```

1 df.head()



	date		datetime	cash_type		card	money	coffee_name	month	day	hour	
0	2024-03-01	2024-03-01	10:15:50.520	card	ANON-0000-0000-0001		38.7	Latte	2024-03	5	10	
1	2024-03-01	2024-03-01	12:19:22.539	card	ANON-0000-0000-0002		38.7	Hot Chocolate	2024-03	5	12	
2	2024-03-01	2024-03-01	12:20:18.089	card	ANON-0000-0000-0002		38.7	Hot Chocolate	2024-03	5	12	
3	2024-03-01	2024-03-01	13:46:33.006	card	ANON-0000-0000-0003		28.9	Americano	2024-03	5	13	
4	2024-03-01	2024-03-01	13:48:14.626	card	ANON-0000-0000-0004		38.7	Latte	2024-03	5	13	



Next steps:

Generate code with df

 View recommended plots

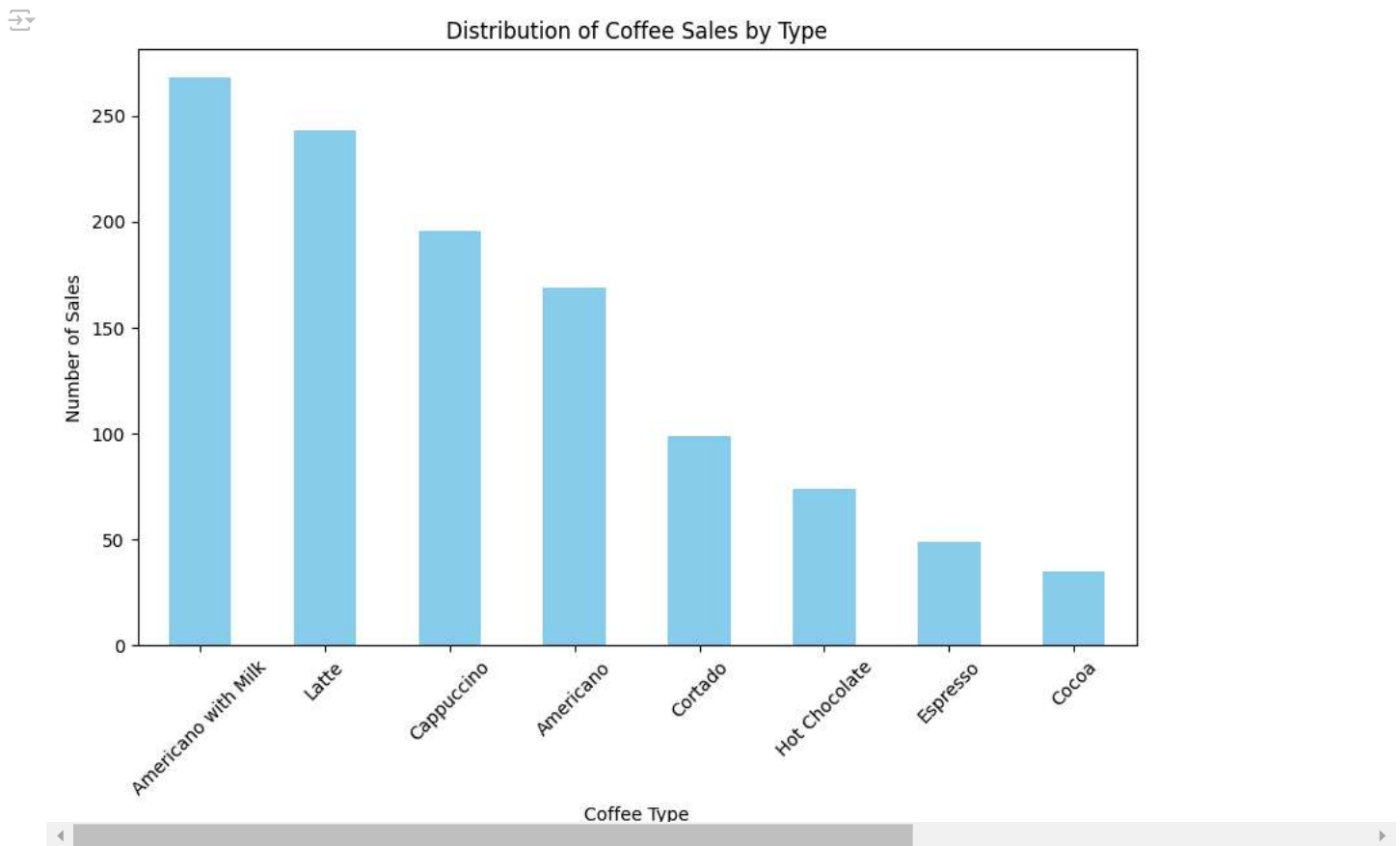
New interactive sheet

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✓ 1. What is the distribution of coffee sales by type?

```
1 coffee_counts = df['coffee_name'].value_counts()
2
3 plt.figure(figsize=(10, 6))
4 coffee_counts.plot(kind='bar', color='skyblue')
5 plt.title('Distribution of Coffee Sales by Type')
6 plt.xlabel('Coffee Type')
7 plt.ylabel('Number of Sales')
8 plt.xticks(rotation=45)
9 plt.show()
```



With the analysis of above graph , we came to the conclusion that "Americano with Milk" has the highest sales

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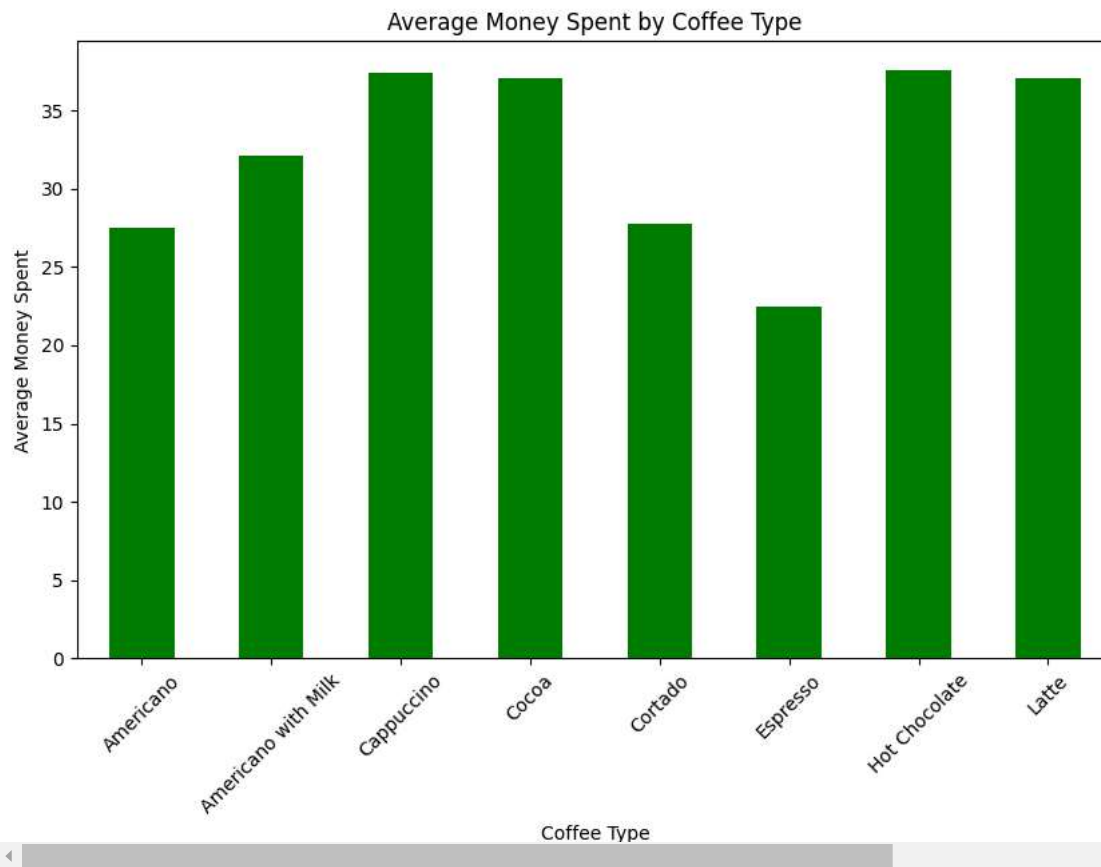
✓ 2. What is the average money spent for each coffee type?

```
1 avg_spent = df.groupby('coffee_name')['money'].mean()
2
3 plt.figure(figsize=(10, 6))
4 avg_spent.plot(kind='bar', color='green')
```

```

5 plt.title('Average Money Spent by Coffee Type')
6 plt.xlabel('Coffee Type')
7 plt.ylabel('Average Money Spent')
8 plt.xticks(rotation=45)
9 plt.show()
10

```



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✓ 3. How does payment method distribution look?

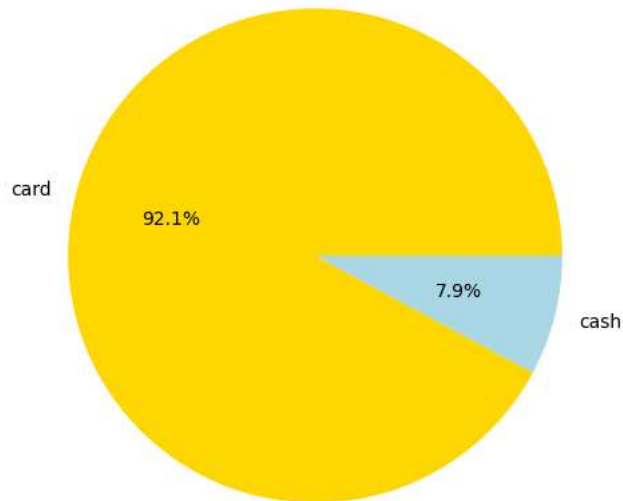
```

1 payment_counts = df['cash_type'].value_counts()
2
3 plt.figure(figsize=(6, 6))
4 payment_counts.plot(kind='pie', autopct='%1.1f%%', colors=['gold', 'lightblue'])
5 plt.title('Payment Method Distribution')
6 plt.ylabel('')
7 plt.show()
8

```



Payment Method Distribution



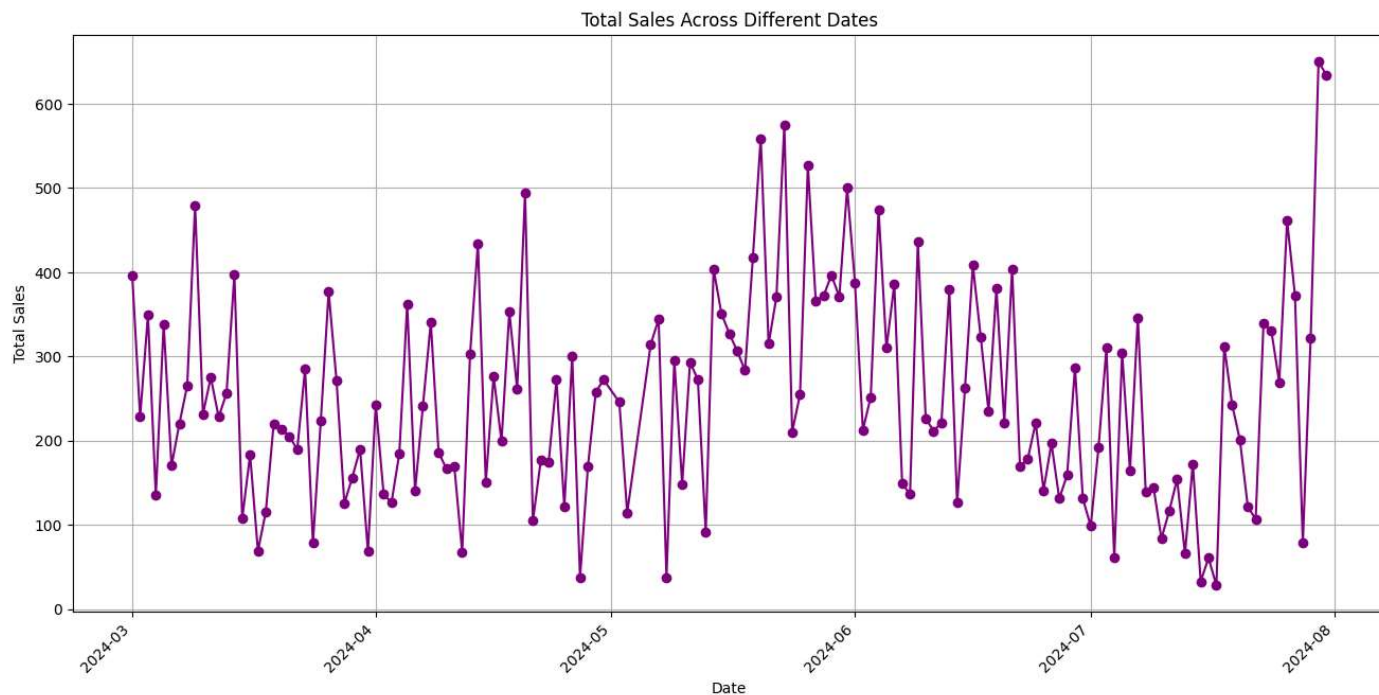
Most payments are done using card

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✓ 4. How do total sales compare across different dates?

```
1 date_sales = df.groupby('date')['money'].sum()
2
3
4 plt.figure(figsize=(16, 8))
5 date_sales.plot(kind='line', marker='o', color='purple')
6 plt.title('Total Sales Across Different Dates')
7 plt.xlabel('Date')
8 plt.ylabel('Total Sales')
9 plt.xticks(rotation=45)
10 plt.grid()
11 plt.show()
12
```

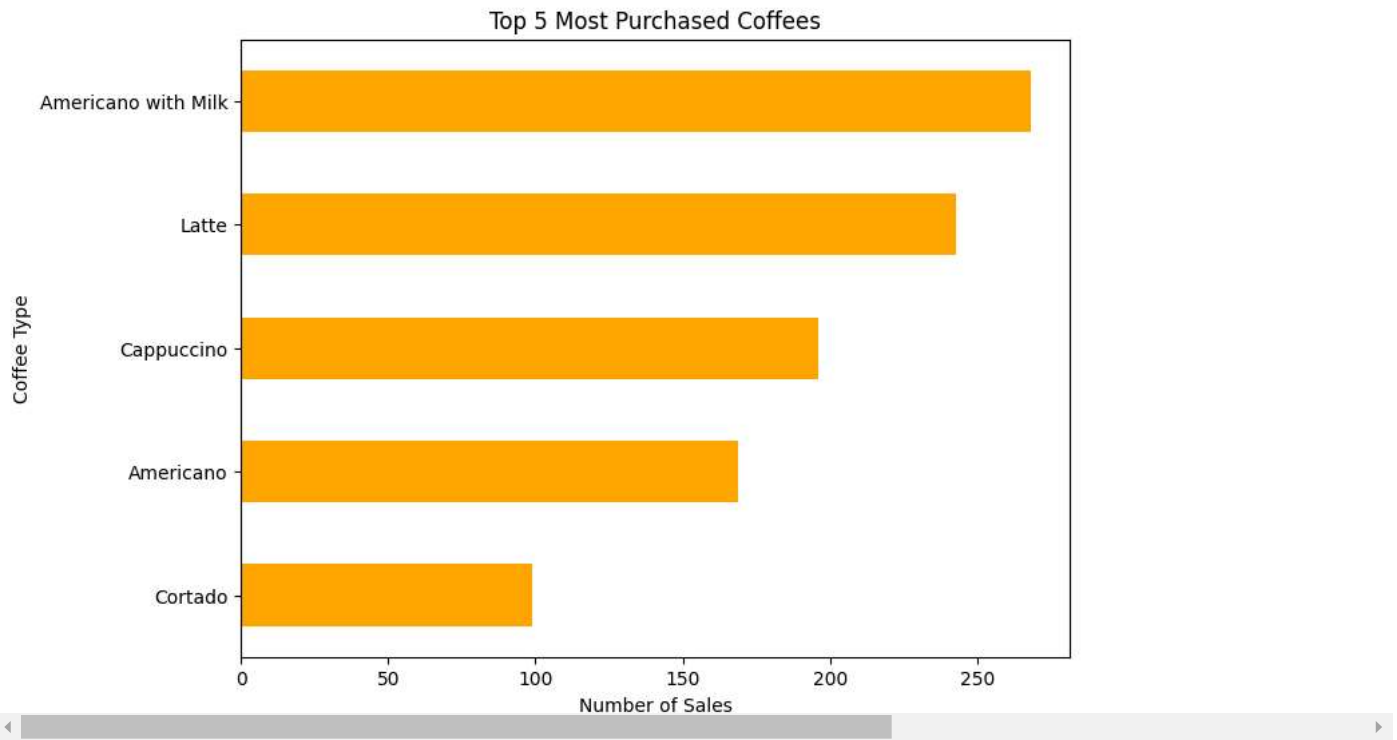


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✓ 5. What are the top 5 most purchased coffees?

```
1 top_5_coffees = coffee_counts.head(5)
2
3 plt.figure(figsize=(8, 6))
4 top_5_coffees.plot(kind='barh', color='orange')
5 plt.title('Top 5 Most Purchased Coffees')
6 plt.xlabel('Number of Sales')
7 plt.ylabel('Coffee Type')
8 plt.gca().invert_yaxis()
9 plt.show()
10
```



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6. What is the total revenue generated by each coffee type?

```
1 revenue = df.groupby(['coffee_name']).sum(['money']).sort_values(by='money', ascending=False)
2 revenue
```



coffee_name	money
Latte	9009.14
Americano with Milk	8601.94
Cappuccino	7333.14
Americano	4644.54
Hot Chocolate	2778.48
Cortado	2745.08
Cocoa	1295.94
Espresso	1100.62

Next steps:

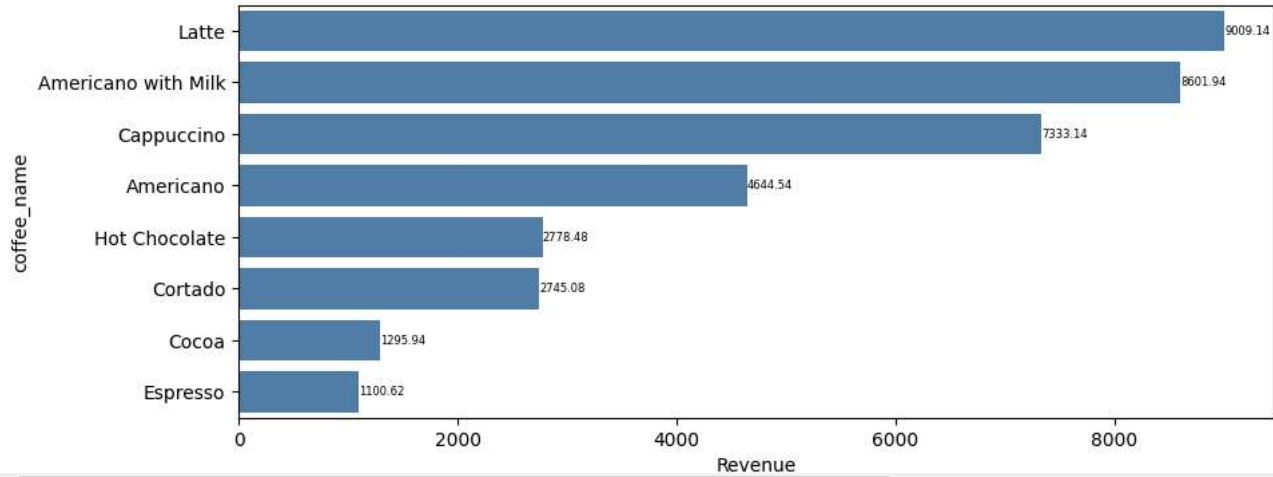
[Generate code with revenue](#)

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[New interactive sheet](#)

```
1 plt.figure(figsize=(10,4))
2 ax = sns.barplot(data=revenue, x='money', y='coffee_name', color='steelblue')
3 ax.bar_label(ax.containers[0], fontsize=6)
4 plt.xlabel("Revenue")
```

```
Text(0.5, 0, 'Revenue')
```

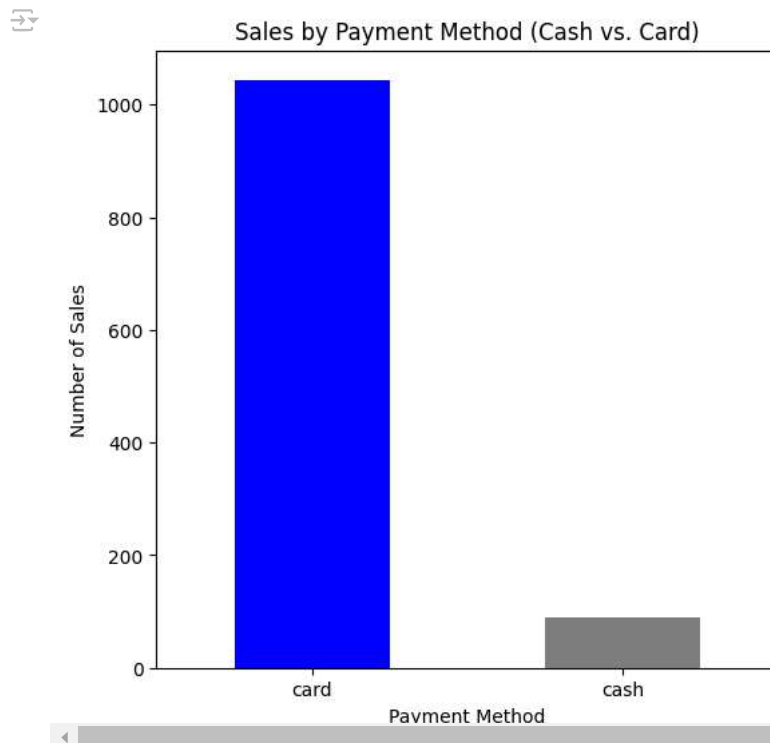


The product with the highest revenue is Americano with Milk whereas Espresso is at the bottom

1 Start coding or generate with AI.

7. How many sales were made for cash vs. card?

```
1 payment_counts = df.groupby('cash_type')['money'].count()
2
3 plt.figure(figsize=(6, 6))
4 payment_counts.plot(kind='bar', color=['blue', 'gray'])
5 plt.title('Sales by Payment Method (Cash vs. Card)')
6 plt.xlabel('Payment Method')
7 plt.ylabel('Number of Sales')
8 plt.xticks(rotation=0)
9 plt.show()
10
```



Payments are done using card with more than 1000 Transactions

```
1 Start coding or generate with AI.

1 Start coding or generate with AI.
```

8. What are the monthly sales for each coffee?

```
1 monthly_sales = df.groupby(['coffee_name', 'month']).count()['date'].reset_index().rename(columns={'date': '
2 monthly_sales
```

coffee_name	month	Americano	Americano with Milk	Cappuccino	Cocoa	Cortado	Espresso	Hot Chocolate	Latte
0	2024-03	36	34	20	6	30	10	22	48
1	2024-04	35	42	43	6	19	7	13	31
2	2024-05	48	58	55	9	17	8	14	58
3	2024-06	14	69	46	5	19	10	14	50
4	2024-07	36	65	32	9	14	14	11	56

Next steps:

Generate code with monthly_sales

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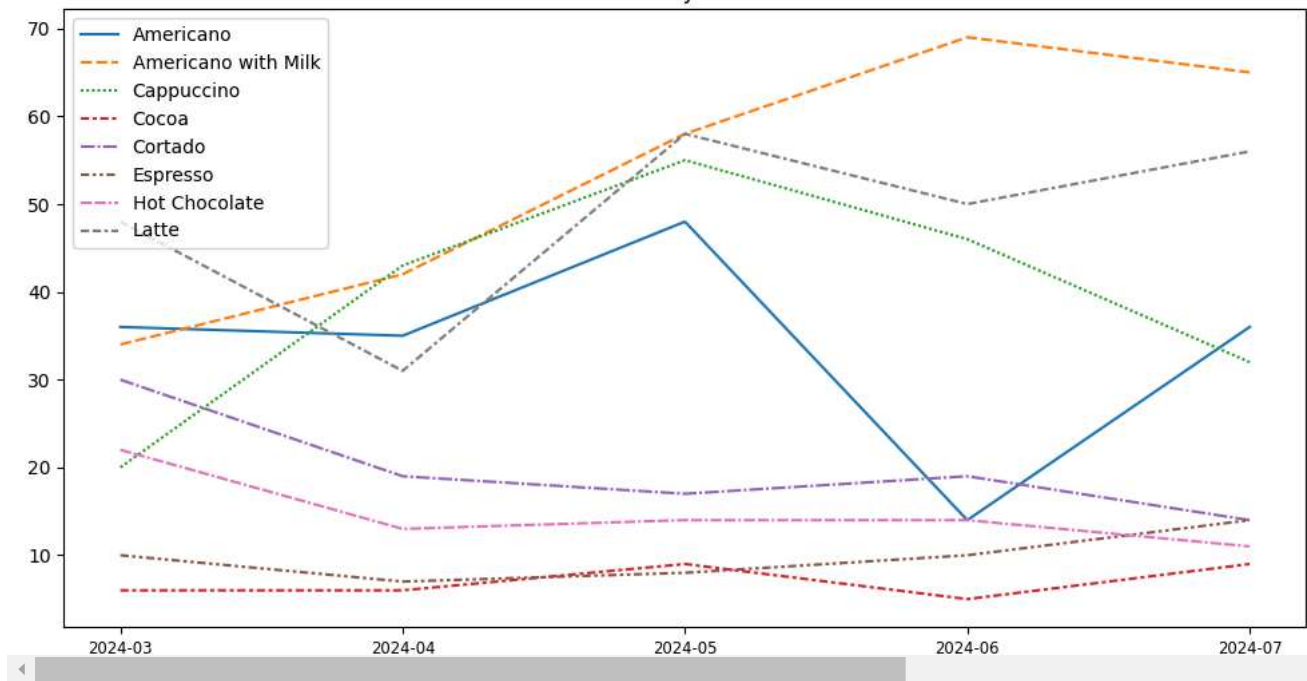
```
1 monthly_sales.describe().T.loc[:,['min','max']]
```

	min	max
coffee_name		
Americano	14.0	48.0
Americano with Milk	34.0	69.0
Cappuccino	20.0	55.0
Cocoa	5.0	9.0
Cortado	14.0	30.0
Espresso	7.0	14.0
Hot Chocolate	11.0	22.0
Latte	31.0	58.0

```
1 plt.figure(figsize=(12,6))
2 sns.lineplot(data=monthly_sales)
3 plt.legend(loc='upper left')
4 plt.title("Monthly Sales")
5 plt.xticks(range(len(monthly_sales['month'])),monthly_sales['month'],size='small')
6 plt.show()
```



Monthly Sales



As shown in the line chart above, Americano with Milk and Latte, and Cappuccino are top selling coffee types, while Cocoa and Espresso have lowest sales. Additionally, Americano with Milk and Latte show an upward trending.

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9. What are the weekday sales for coffee?

1 Start coding or generate with AI.

```
1 weekday_sales = df.groupby(['day']).count()['date'].reset_index().rename(columns={'date': 'count'})
2 weekday_sales
```



	day	count
0	0	151
1	1	151
2	2	185
3	3	165
4	4	164
5	5	163
6	6	154



Next steps:

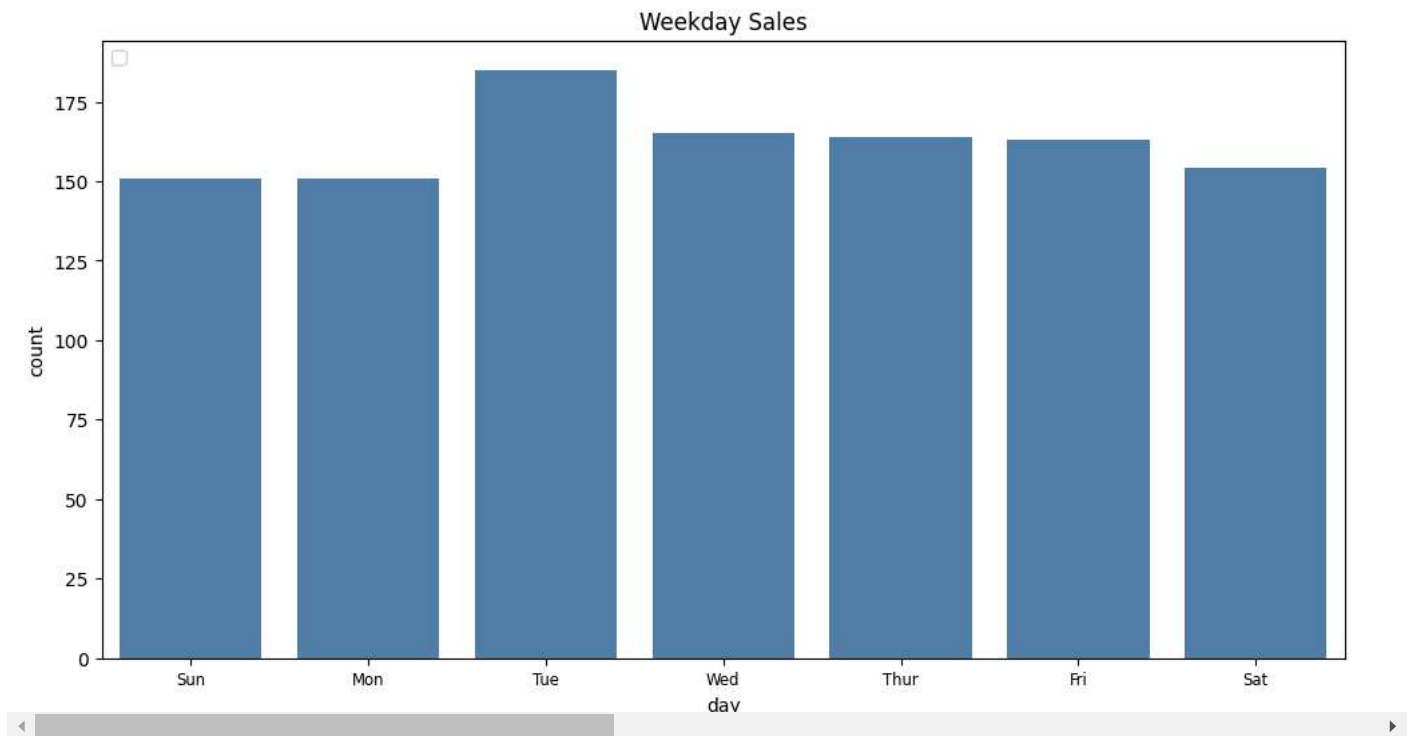
[Generate code with weekday_sales](#)

[View recommended plots](#)

[New interactive sheet](#)

```
1 plt.figure(figsize=(12,6))
2 sns.barplot(data=weekday_sales, x='day' , y='count', color='steelblue')
3 plt.legend(loc='upper left')
4 plt.title("Weekday Sales")
5 plt.xticks(range(len(weekday_sales['day'])),['Sun','Mon','Tue','Wed','Thur','Fri','Sat'],size='small')
6 plt.show()
```

 <ipython-input-30-0436132b0ee6>:3: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with plt.legend(loc='upper left')



The bar chart reveals that Tuesday has the highest sales of the week, while sales on the other days are relatively similar.

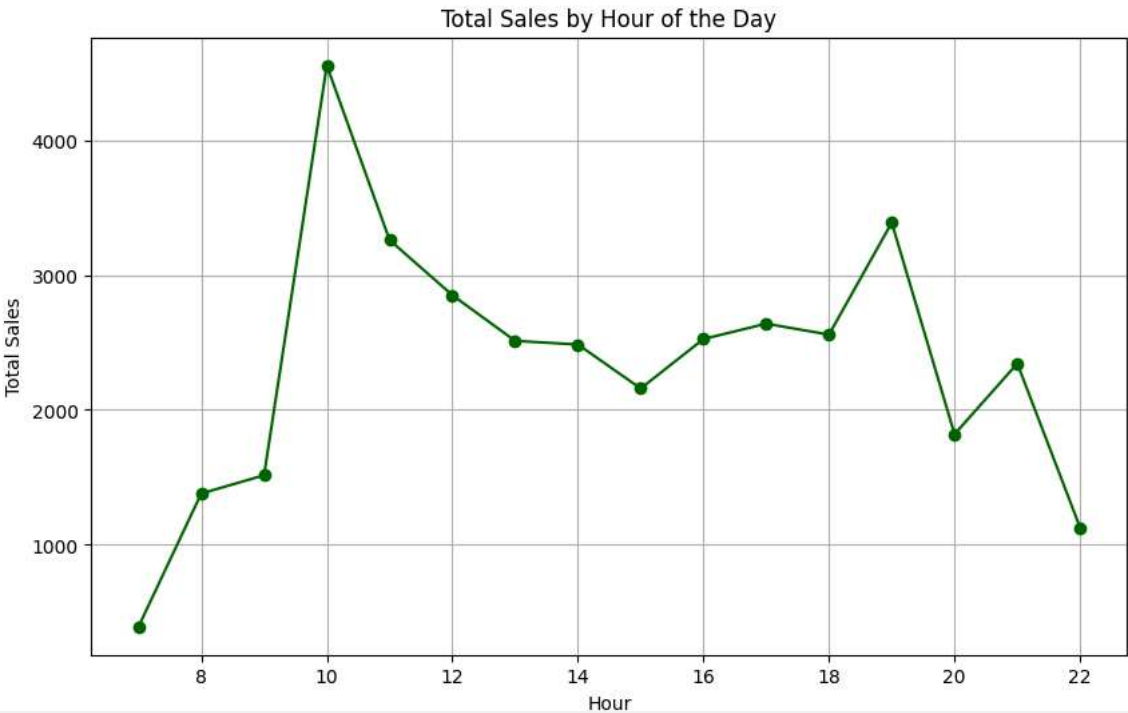
1 Start coding or generate with AI.

✓ 10. What are the total sales by hour of the day?

```

1 df['hour'] = pd.to_datetime(df['datetime'], errors='coerce').dt.hour
2 hourly_sales = df.groupby('hour')['money'].sum()
3
4
5 plt.figure(figsize=(10, 6))
6 hourly_sales.plot(kind='line', marker='o', color='darkgreen')
7 plt.title('Total Sales by Hour of the Day')
8 plt.xlabel('Hour')
9 plt.ylabel('Total Sales')
10 plt.grid()
11 plt.show()
12

```



The plots above illustrate the shopping traffic for each product throughout the day. Notably, all products experience a peak in traffic around 10:00 AM, with this trend being particularly pronounced for Latte. Additionally, Cappuccino, Cocoa, and Hot Chocolate tend to be more popular during the evening hours, specifically between 6:00pm and 8:00pm.

1 Start coding or generate with AI.

```
1 daily_sales = df.groupby(['coffee_name', 'date']).count()['datetime'].reset_index().reset_index().rename(cc
2 daily_sales
```



coffee_name	date	Americano	Americano with Milk	Cappuccino	Cocoa	Cortado	Espresso	Hot Chocolate	Latte
0	2024-03-01	1.0	4.0	0.0	1.0	0.0	0.0	3.0	2.0
1	2024-03-02	3.0	3.0	0.0	0.0	0.0	0.0	0.0	1.0
2	2024-03-03	1.0	2.0	0.0	1.0	2.0	0.0	2.0	2.0
3	2024-03-04	0.0	1.0	0.0	0.0	0.0	1.0	0.0	2.0
4	2024-03-05	0.0	0.0	0.0	1.0	1.0	0.0	4.0	3.0
...	...	...	...	...	...	...	...	...	...
145	2024-07-27	0.0	5.0	4.0	0.0	0.0	2.0	0.0	2.0
146	2024-07-28	0.0	1.0	0.0	0.0	0.0	1.0	0.0	1.0
147	2024-07-29	3.0	2.0	2.0	1.0	0.0	0.0	2.0	1.0
148	2024-07-30	2.0	12.0	2.0	0.0	3.0	2.0	0.0	3.0
149	2024-07-31	2.0	6.0	1.0	2.0	4.0	0.0	0.0	7.0

150 rows × 9 columns

Next steps:

Generate code with daily_sales

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```
1 daily_sales.iloc[:,1:].describe().T.loc[:,['min','max']]
```



	min	max
coffee_name		
Americano	0.0	5.0
Americano with Milk	0.0	12.0
Cappuccino	0.0	9.0
Cocoa	0.0	2.0



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